Quantic School of Business & Technology

Managing Software Engineering

MSSE Managing Software Engineering Project

Vishant Prasad
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MSSE – Quantic School of Business & Technology

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Client/Seller: Lily - Florist

4 Requirements Gathering Discussion

Key Takeaways

The link to the **ChatGPT** interview with the Lily Florist AI Agent can be found here:

https://chatgpt.com/share/68c83ff5-68c0-800b-b3be-10856c47a4b3

A copy of the interview exported has also been attached at the end of this document.

4.1 Objective

The main purpose is to make it easy for customers to order flowers online, explore my designs and schedule deliveries or pickups without hassle. For me (Lily), it helps manage orders and inventory more efficiently so I can focus on creating arrangements and serving customers better.

4.2 Products Sold

1. Fresh Flower Arrangements
    a. Bouquets
    b. Centrepieces
    c. Seasonal Designs
2. Plants
3. Gift Baskets
4. Small Add-Ons;
    a. Candles
    b. Chocolates
5. Flowers for Services such as;
    a. Weddings
    b. Parties
    c. Corporate Functions

4.3 Pain Points

Table 1: Pain Points from Client

ID	Category	Description	Theme
P1	Ordering & Confirmation Issues	Customers sometimes feel unsure if their order went through properly.	Lack of Transparency & Confirmation in Ordering Process
P2	Delivery & Time Sensitivity	Delivery timing can cause stress, especially for last-minute orders or surprise gifts.	Scheduling, Timeliness & Reliability Concerns
P3	Product Representation & Expectations	People want to see clear images of arrangements, but occasionally the flowers in stock differ from the photos.	Accuracy of Product Representation vs Real Availability
P4	Customer Convenience & Personalisation	Repeating past orders isn't always easy, so customers have to explain their preferences again.	Customer History, Personalisation &

			Ease of Repeat Ordering

4.4 Website Features

Table 2: Website Features from Client

Feature	Description	User Role
Online Ordering	Enables customers to place orders for floral arrangements with an easy checkout process.	Customer
Gallery	Showcases floral arrangement for customers to browse and choose designs.	Customer
Seasonal Updates & Blog	Provides flower tips, seasonal inspiration and stories to engage customers.	Customer
Inventory Tracking	Displays available stock so customers can only order items in stock.	Customer/Seller
Delivery Scheduling	Allows customers to select delivery date and time options for their orders.	Customer
Customer Accounts	Let's customers create accounts, view order history and easily repeat past purchases.	Customer
Social Media Links	Connects customers to social media channels to stay updated and engaged.	Customer/Seller

4.5 Target Goals

Table 3: Target Goals

Category	Target Goals
Business Growth	- Increase online orders with a smooth checkout process. - Showcase my floral designs in a beautiful and inspiring manner. - Share seasonal updates and flower care tips through a blog.
Customer Experience	- Improve customer experience with accounts for repeat purchases. - Offer flexible delivery scheduling for customers.
Operational Efficiency	- Manage inventory to avoid shortages or overstock.

5 User Stories

5.1 Draft User Stories

The following are categorised draft user stories from the interview with Lily:

1. **Ordering & Confirmation Issues**
    a. ***Pain Point:*** Customers sometimes feel unsure if their order went through properly when placed online or by phone.
    b. ***Theme:*** Lack of transparency/confirmation in the ordering process.
2. **Delivery & Time Sensitivity**

a. ***Pain Point:*** Delivery timing can cause stress, especially for last-minute orders or surprise gifts.
b. ***Theme:*** Scheduling, timeliness and reliability concerns.

3. **Product Representation & Expectations**
    a. ***Pain Point:*** People want to see clear images of arrangements, but occasionally the flowers in stock differ from the photos.
    b. ***Theme:*** Accuracy of the product representation vs. real availability.
4. **Customer Convenience & Personalisation**
    a. ***Pain Points:*** Repeating past orders isn't always easy, so customers have to explain their preferences again.
    b. ***Theme:*** Customer history, personalisation and ease of repeat ordering.

5.2 List of User Stories

5.2.1 Customer-Facing Stories

1. **US1**: As a customer, I want to browse a gallery of floral arrangements so that I can choose designs I like.
2. **US2**: As a customer, I want to read seasonal updates and flower tips so that I feel inspired and informed when making a purchase.
3. **US3**: As a customer, I want to see what arrangements are currently in stock so that I only order items available for delivery.
4. **US4**: As a customer, I want an easy checkout process so that I can complete orders quickly without confusion and with a clear pricing.
5. **US5**: As a customer, I want to pay securely using cards or digital wallets.
6. **US6**: As a customer, I want to schedule a delivery date and time so that my flowers arrive when I need them.
7. **US7**: As a customer, I want to create an account and view my order history so that I can reorder past arrangements easily.
8. **US8**: As a customer, I want to save favourite arrangements so that I can quickly find and order them later.
9. **US9**: As a customer, I want clear confirmation of my order via email or SMS so that I know my order was successful.
10. **US10**: As a customer, I want notifications when my delivery is out for delivery or completed so that I can track my gift.
11. **US11**: As a customer, I want to have the opportunity to redeem discounts on future orders from points earned.

5.2.2 Florist/Admin Stories

12. **US12**: As the florist, I want to manage inventory in real time so that customers only see available arrangements.
13. **US13**: As the florist, I want to track and manage all incoming orders so that I can efficiently prepare and schedule deliveries.
14. **US14**: As the florist, I want to update the gallery and blog content easily so that customers see fresh designs and seasonal tips.
15. **US15**: As the florist, I want to manage delivery scheduling so that I can plan routes efficiently and avoid conflicts.
16. **US16**: As the florist, I want to maintain customer accounts and preferences so that repeat customers get a personalised experience.

17. **US17**: As the florist, I want to provide links to social media so that customers can follow updates and promotions.
18. **US18**: As the florist, I want to provide points and rewards as part of a rewards program for referrals and special promotions during holidays and events.
19. **US19**: As the florist, I want to promote and refer people to some of the shop's profile pages on social media platforms Instagram and Facebook.

6 User Journey

6.1 User Goal

Selected User Story: *"As a customer, I want to order flowers online and schedule a delivery so that my gift arrives on time."*

Key Goals:

- Order Flowers Online
- Schedule a Delivery

Table 4: Objectives & Key Goals

Target Goal	Related Feature	Related User Story(s)
Increase Online Orders	Online Ordering	4, 5
Showcase Designs	Gallery	1
Flexible Delivery	Delivery Scheduling	6
Manage Inventory	Inventory Tracking	12
Improve Customer Experience	Customer Accounts	7, 8
Share Seasonal Updates	Seasonal Updates & Blogs	2

6.2 Breakdown of User Journey

Table 5: User Journey

Step	User Role	User Action	System Response
1	Customer	Visits the florist's **website**.	Homepage loads with navigation to gallery, blog and ordering options.
2	Customer	Browses the **gallery** of arrangements	System displays designs with clear images, prices and availability.
3	Customer	Selects a floral arrangement.	System shows product details, stock status and customisation options (e.g. note card).
4	Customer	Adds the arrangement to the **cart**.	System updates cart and shows subtotal.
5	Customer	Proceeds to **checkout**.	System prompts for login/guest checkout.
6	Customer	**Logs in** (or **creates** an **account**).	System retrieves stored addresses, past orders and payment details.
7	Customer	Enters **delivery address** and selects **date/time slot**.	System validates availability and confirms delivery slot.
8	Customer	Provides **payment details** and places **order**.	System securely processes payment.

9	Customer	Receives **order confirmation** on screen by **email** and/or **SMS**.	Order saved, florist notified of incoming order.
10	Florist	**Reviews** new order in **backend dashboard**.	System shows customer info, delivery time and arrangement details.
11	Florist	Prepares flowers and arranges **delivery**.	System updates status (e.g. "in preparation", "out for delivery").
12	Customer	Receives **notifications** (out for delivery, delivered).	System ensures customer is informed until delivery completion.

See Appendix for visualisation via User Journey Flow Diagram - Figure 1 and - Figure 2.

7 Requirements Specification

7.1 Introduction

The following section defines the functional and non-functional requirements for Lily's Florist Shop and the online ordering system based off the aforementioned User Journey that has been generated. The system enables customers to browse, select and order floral arrangements while providing inventory, delivery and account management features for the florist. The following requirements are derived from user stories, pain points and target goals documented above.

7.2 Functional Requirements (FR)

Table 6: Functional Requirements

Req. ID	Category	Requirement	Related User Story(s)
FR1	Ordering	A customer shall be able to browse a gallery of floral arrangements with images, prices and stock availability.	US1
FR2	Ordering	A customer shall be able to add a selected arrangement to a shopping cart.	US4
FR3	Ordering	A customer shall be able to remove or modify items in their cart before checkout.	US4
FR4	Checkout	A customer shall be able to proceed to checkout as a guest or logged-in user.	US4 & US5
FR5	Login	A user shall be allowed three login attempts before the system locks the account for 30 minutes.	US5 & US7
FR6	Payment	A customer shall be able to pay securely using credit/debit cards or digital wallets.	US5
FR7	Delivery	A customer shall be able to select a delivery date and time from available slots.	US6
FR8	Confirmation	The system shall provide a confirmation of order on-screen and via email/SMS immediately after payment.	US9
FR9	Notification	Customers shall receive notifications when the order status (e.g. in preparation, out for delivery, delivered).	US10 & US13

FR10	Account Management	Customers shall be able to create accounts, view order history and reorder past arrangements.	US7
FR11	Favourites	Customers shall be able to save favourite arrangements to quickly find and reorder them later.	US7 & US8
FR12	Inventory	The florist shall be able to update inventory in real-time to reflect available arrangements.	US12
FR13	Order Management	The florist shall be able to view all incoming orders and track status in a dashboard.	US13
FR14	Content Management	The florist shall be able to update gallery and blog content through an admin interface.	US14
FR15	Delivery Planning	The florist shall be able to manage delivery schedules and plan routes efficiently.	US15
FR16	Customer Preferences	The system shall store customer preferences to allow personalised experiences for repeat customers.	US16
FR17	Social Media Integration	The system shall provide links to social media platforms (Instagram, Facebook) for promotions.	US17 & US19
FR18	Rewards Program	The system shall allow points and rewards redemption for referrals and special promotions.	US11 & US18
FR19	Product Details	System shall display accurate product details including stock availability, price and customisation options.	US3
FR20	Blogs & Tips	Customers shall be able to view seasonal updates and flower care tips in a blog section.	US2
FR21	Search & Filtering	Customers shall be able to filter gallery items by type, occasion and availability.	US1
FR22	Multi-Device Access	Customers shall be able to access the website from desktop and mobile devices with consistent functionality.	US1 & US4

7.3 Non-Functional Requirements (NFR)

Table 7: Non-Functional Requirements

Req. ID	Category	Requirement	Related FR/User Story
NFR1	Performance	Pages shall load within 3 seconds for users on standard broadband connections.	FR1 & FR2
NFR2	Security	All payment transactions shall comply with PCI DSS standards.	FR6
NFR3	Security	Customer passwords shall be stored using salted hashing algorithms.	FR5
NFR4	Availability	System shall have 99% uptime during business hours (8AM – 8PM).	FR1 – FR22
NFR5	Data Retention	Customer data shall be stored for a minimum of 5 years if no activity occurs on the account.	FR10 & FR16
NFR6	Usability	The website shall be navigable with standard screen sizes (desktop, tablet, mobile).	FR22
NFR7	Scalability	System shall support at least 500 concurrent users without performance degradation.	F1 – F22

NFR8	Backup	Daily automated backups shall be maintained for database and media files.	F12 & F13
NFR9	Compliance	Emails/SMS notifications shall comply with privacy regulations.	FR8 & FR9
NFR10	Logging	All transactions and order status updates shall be logged with timestamps for audit purposes.	FR6 & FR13

8 Website Wireframe

8.1 Site Map

Below is the illustrated site map of the potential website that would satisfy the user journey and requirements that have been documented under headings 6 User Journey and 7 Requirements Specification.

Based off this site map the website wireframes have been created. See Appendix section 9.3 Wireframes for the individual wireframes without annotations of each of the web pages documented in the site map as well as participant or admin only viewable web pages.

Annotated versions of the wireframes are documented under this section under the heading 8.2 Wireframes Annotated.

A mockup of each of the web pages developed from each of the wireframes created are also documented in the Appendix under section 9.4 Mock-ups.

Images merged on the next page for the site map diagrams.

Note: Certain pages of the website have not had wireframe or mock-ups generated as they are not a part of the selected User Journey. These include:

- About Page
- Blog Page
- FAQs Page
- Privacy Policy Page
- Reviews View
- Form Submission Animations (Contact Us)
- Administrator Dashboard Settings Tab
- Administrator Dashboard Export CSV Interface
- Administrator Refund Interface
- Forgot Password Process